

LUMINA-2M as an AC Power-Flow Surrogate

Exact formulation of the released HGT and PPC adaptation

Implementation note for `train_valid_test_lumina.py`

August 2026

Scope. This document describes the released LUMINA Heterogeneous Graph Transformer loaded from `argonne/LUMINA-2M`, adapted to the PPC AC-PF parquets. LUMINA was pretrained for AC-OPF-style heterogeneous inputs; here its bus voltage output is fine-tuned or retrained against Newton PF solutions. Optional known-operator postprocessing is excluded.

1 AC-PF Target

For voltage $\underline{V}_i = v_i e^{j\theta_i}$,

$$\underline{\mathbf{S}}^{\text{calc}}(\mathbf{v}, \boldsymbol{\theta}) = \underline{\mathbf{V}} \odot \overline{\mathbf{Y}\underline{\mathbf{V}}}. \quad (1)$$

The mismatch is

$$\Delta \mathbf{P} = \mathbf{P}^{\text{set}} - \Re \underline{\mathbf{S}}^{\text{calc}}, \quad \Delta \mathbf{Q} = \mathbf{Q}^{\text{set}} - \Im \underline{\mathbf{S}}^{\text{calc}}. \quad (2)$$

The adapted LUMINA model learns

$$f_{\Theta}(\mathcal{G}, \underline{\mathbf{S}}^{\text{set}}) \mapsto (\hat{\mathbf{v}}, \hat{\boldsymbol{\theta}}) \approx (\mathbf{v}^*, \boldsymbol{\theta}^*). \quad (3)$$

2 Adapted Heterogeneous Graph

LUMINA uses four node types:

$$\mathcal{V} = \mathcal{V}_b \cup \tilde{\mathcal{V}}_g \cup \mathcal{V}_d \cup \mathcal{V}_s, \quad (4)$$

representing buses, proxy generators, loads, and shunts. It uses eight directed relation types:

$$b \xrightarrow{\text{ac line}} b, \quad b \xrightarrow{\text{transformer}} b, \quad (5)$$

$$\tilde{g} \xrightleftharpoons{\text{generator link}} b, \quad d \xrightleftharpoons{\text{load link}} b, \quad s \xrightleftharpoons{\text{shunt link}} b. \quad (6)$$

The source LUMINA schema supports physical generator nodes. Since the PPC PF parquet is bus-aggregated, the adapter constructs one proxy generator on every PV/REF bus.

2.1 Node features

The bus row is

$$\mathbf{x}_i^b = [V_{n,i}^{\text{kV}}, v_i^{\min}, v_i^{\max}, \mathbb{K}_{\text{PQ}}, \mathbb{K}_{\text{PV}}, \mathbb{K}_{\text{REF}}, \mathbb{K}_{\text{ISO}}] \in \mathbb{R}^7. \quad (7)$$

The generator, load, and shunt rows are

$$\mathbf{x}_{\tilde{g}_i}^g = [m_{\text{base}}, P_g, P_g^{\min}, P_g^{\max}, Q_g, Q_g^{\min}, Q_g^{\max}, v_g^{\text{set}}, c_2, c_1, c_0] \in \mathbb{R}^{11}, \quad (8)$$

$$\mathbf{x}_{d_i}^d = [P_{d,i}, Q_{d,i}] \in \mathbb{R}^2, \quad (9)$$

$$\mathbf{x}_{s_i}^s = [B_i^{\text{sh}}, G_i^{\text{sh}}] \in \mathbb{R}^2. \quad (10)$$

In the PF adaptation,

- P_g, Q_g may be filled with the known bus-level injection (`gen_setpoint_mode=known`);
- generator limits are synthesized around that injection and costs are zero;
- $v_g^{\text{set}} = v_i^0$ at proxy-generator buses;
- load channels are the clipped negative net injection;
- the unspecified Q slot at PV/REF buses is zeroed before graph construction.

The bus row itself does *not* include (v_i^0, θ_i^0) . Only the generator setpoint channel exposes v_i^0 at PV/REF buses. Therefore the manual-flat versus DC-start angle is not directly provided to released LUMINA.

2.2 Branch attributes supplied by the adapter

AC-line and transformer rows follow the PGLib/OPF schema:

$$\mathbf{e}_{ij}^\ell = [\theta^{\min}, \theta^{\max}, b_{fr}, b_{to}, r, x, S_A, S_B, S_C] \in \mathbb{R}^9, \quad (11)$$

$$\mathbf{e}_{ij}^\tau = [\theta^{\min}, \theta^{\max}, r, x, S_A, S_B, S_C, \tau, \phi, b_{fr}, b_{to}] \in \mathbb{R}^{11}. \quad (12)$$

They are reconstructed from PPC branch admittance, tap, phase shift, and shunts.

Implementation-critical fact. The released `HGT.forward` accepts an `edge_attr_dict` argument but does not pass it into `HGTConv`. Consequently, the present LUMINA model uses branch attributes only indirectly through graph construction/classification; the numerical values (r, x, b, τ, ϕ) do not enter message attention. The relations distinguish “AC line” from “transformer”, but two lines with the same endpoints and different impedance are indistinguishable to the HGT core.

3 Type Projections

Each node type $t \in \{b, g, d, s\}$ has its own input projection:

$$\mathbf{h}_i^{t,(0)} = \text{ReLU}(W_{\text{in}}^t \mathbf{x}_i^t + \mathbf{b}_{\text{in}}^t). \quad (13)$$

The released checkpoint configuration uses

$$d = 128, \quad L = 6, \quad H = 2, \quad p_{\text{drop}} = 0.2439835578. \quad (14)$$

4 Heterogeneous Graph Transformer Layers

Let relation $r = (s, \varrho, t)$ connect source type s to target type t . For head h , HGT forms type-specific queries, keys, and values:

$$\mathbf{q}_i^{t,h} = W_Q^{t,h} \mathbf{h}_i^t, \quad \mathbf{k}_j^{s,h} = W_K^{s,h} \mathbf{h}_j^s, \quad \mathbf{v}_j^{s,h} = W_V^{s,h} \mathbf{h}_j^s. \quad (15)$$

The relation transforms key/value information and supplies a learned relation prior. An operator-level form is

$$e_{ji}^{r,h} = \frac{(\mathbf{q}_i^{t,h})^\top A_r^h \mathbf{k}_j^{s,h}}{\sqrt{d_h}} + \mu_r^h, \quad (16)$$

$$\alpha_{ji}^{r,h} = \text{softmax}_{(r',u):u \rightarrow i}(e_{ui}^{r',h}), \quad (17)$$

$$\mathbf{m}_i^{t,h} = \sum_{r=(s,\cdot,t)} \sum_{j \in \mathcal{N}_r(i)} \alpha_{ji}^{r,h} M_r^h \mathbf{v}_j^{s,h}. \quad (18)$$

Thus different relations learn different transformations even when they share a target node type. The heads are combined, transformed by a target-type output map, and mixed with a type-specific skip path:

$$\mathbf{h}_i^{t,(\ell+1)} = \text{ReLU} \left(\text{HGTConv}_t^{(\ell)}(\{\mathbf{h}^{(\ell)}\}, \{\mathcal{E}_r\})_i \right). \quad (19)$$

Dropout follows the input projection and every HGT layer during training.

Unlike GraphKit’s HGNS, LUMINA has no layerwise voltage decoding, no branch-flow calculation, and no AC residual feedback inside its forward pass.

5 Terminal Output Heads

Only after the sixth HGT layer, two type-specific linear heads are applied:

$$[z_{\theta,i}, z_{v,i}] = W_{\text{out}}^b \mathbf{h}_i^{b,(L)} + \mathbf{b}_{\text{out}}^b, \quad (20)$$

$$[z_{P,g}, z_{Q,g}] = W_{\text{out}}^g \mathbf{h}_g^{g,(L)} + \mathbf{b}_{\text{out}}^g. \quad (21)$$

With min-max scaling enabled, the magnitude and generator outputs are bounded:

$$\hat{v}_i = v_i^{\min} + (v_i^{\max} - v_i^{\min})\sigma(z_{v,i}), \quad (22)$$

$$\hat{P}_g = P_g^{\min} + (P_g^{\max} - P_g^{\min})\sigma(z_{P,g}), \quad (23)$$

$$\hat{Q}_g = Q_g^{\min} + (Q_g^{\max} - Q_g^{\min})\sigma(z_{Q,g}). \quad (24)$$

The angle is not bounded by min-max scaling; the adapter wraps it:

$$\hat{\theta}_i = \text{atan2}(\sin z_{\theta,i}, \cos z_{\theta,i}). \quad (25)$$

The AC-PF path consumes

$$\hat{\mathbf{V}}_{\text{state}} = \{(\hat{v}_i, \hat{\theta}_i)\}_{i=1}^N. \quad (26)$$

The generator output head is unused under the default voltage-only PF objective. The forward function also does not pin known PF quantities: it predicts magnitude at PV/REF buses and angle at the REF bus, and supervision must learn those identities.

6 External AC-PF Physics Loss

The supervised term is

$$\mathcal{L}_{\text{state}} = \frac{1}{N} \sum_i \left[(\hat{v}_i - v_i^*)^2 + \text{wrap}(\hat{\theta}_i - \theta_i^*)^2 \right]. \quad (27)$$

Let

$$\mathcal{M}_P = \mathcal{V}_b \setminus \mathcal{V}_{\text{REF}}, \quad \mathcal{M}_Q = \mathcal{V}_{\text{PQ}}. \quad (28)$$

The external physics term uses the exact PPC $\mathbf{Y}$, independent of the HGT edge representation:

$$\mathcal{L}_{\text{phys}} = \frac{1}{|\mathcal{M}_P| + |\mathcal{M}_Q|} \left[\sum_{i \in \mathcal{M}_P} \rho(\Delta P_i) + \sum_{i \in \mathcal{M}_Q} \rho(\Delta Q_i) \right], \quad (29)$$

with default

$$\rho(r) = \log \cosh(\text{clip}(r, -30, 30)). \quad (30)$$

The standard objective is

$$\boxed{\mathcal{L} = \mathcal{L}_{\text{state}} + 10^{-2} \mathcal{L}_{\text{phys}}}. \quad (31)$$

Therefore LUMINA can still learn impedance-sensitive AC behavior through the external loss gradient, even though impedance is absent from its message features. It must infer much of that behavior from topology, relation type, nominal voltage, injections, and repeated-grid training examples.

Training physics is evaluated through complex64 casts in the script; reported validation/test residual metrics use the complex128 dataset/Ybus path.

7 Interpretive Summary

Property	LUMINA PF adaptation
Graph	Bus, proxy-generator, load, and shunt nodes with eight directed relation types.
Electrical edge encoding	Line/transformer attributes are constructed, but the released HGT forward currently ignores numerical edge attributes.
Neural core	Six HGTCConv layers, hidden width 128, two heads, and dropout about 0.244.
Voltage decoder	One terminal bus head; magnitude sigmoid-scaled to the voltage band and angle wrapped.
PF constraints inside forward	No PQ/PV/REF pinning, no branch-flow operator, and no residual feedback.
Physics outside forward	Exact-Ybus robust AC-PF loss and complex128 evaluation metrics.
Start-state use	Start magnitude appears only as proxy-generator v_g^{set} ; start angle is omitted.
Pretraining	Released LUMINA-2M weights may be loaded and fine-tuned.

Code provenance

The formulation follows `lumina_inference/model/hetero_model.py`, the released `lumina_config.json`, and the AC-PF adapter in `train_valid_test_lumina.py`, as present in the local workspace in August 2026.